$$\vec{a} = a_1\hat{i}$$
$$\vec{b} = b_1\hat{i} + b_2\hat{j}$$
$$\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$$

Now, $\vec{a}\times(\vec{b}\times\vec{c}) = a_1\hat{i}\times\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ b_1 & b_2 & 0\\ c_1 & c_2 & c_3\end{vmatrix}$

$$= a_1\hat{i}\times\left(b_2c_3\hat{i} - b_1c_3\hat{j} + (b_1c_2 - b_2c_1)\hat{k}\right)$$
$$= -a_1b_1c_3\hat{k} + a_1(b_2c_1 - b_1c_2)\hat{j} \quad \text{... (1)}$$

$$(\vec{a}.\vec{c})\vec{b} - (\vec{a}.\vec{b})\vec{c} = a_1c_1\times\left(b_1\hat{i} + b_2\hat{j}\right) - a_1b_1\left(c_1\hat{i} + c_2\hat{j} + c_3\hat{k}\right)$$
$$= a_1(b_2c_1 - b_1c_2)\hat{j} - a_1b_1c_3\hat{k} \quad \text{... (2)}$$

From equations (1) and (2), we get

$$\vec{a}\times(\vec{b}\times\vec{c}) = (\vec{a}\cdot\vec{c})\vec{b} - (a\cdot\vec{b})\vec{c}$$

Note

(1) $\vec{a}\times(\vec{b}\times\vec{c}) = \alpha\vec{b} + \beta\vec{c}$, where $\alpha = \vec{a}\cdot\vec{c}$ and $\beta = -(\vec{a}\cdot\vec{b})$, and so it lies in the plane parallel to $\vec{b}$ and $\vec{c}$.

(2) We also note that

$$(\vec{a}\times\vec{b})\times\vec{c} = -\vec{c}\times(\vec{a}\times\vec{b})$$
$$= -[(\vec{c}\cdot\vec{b})\vec{a} - (\vec{c}\cdot\vec{a})\vec{b}]$$
$$= (\vec{a}\cdot\vec{c})\vec{b} - (\vec{b}\cdot\vec{c})\vec{a}$$

Therefore, $(\vec{a}\times\vec{b})\times\vec{c}$ lies in the plane parallel to $\vec{a}$ and $\vec{b}$.

(3) In $(\vec{a}\times\vec{b})\times\vec{c}$, consider the vectors inside the brackets, call $\vec{b}$ as the middle vector and $\vec{a}$ as the non-middle vector. Similarly, in $\vec{a}\times(\vec{b}\times\vec{c})$, $\vec{b}$ is the middle vector and $\vec{c}$ is the non-middle vector. Then we observe that a vector triple product of these vectors is equal to

λ (middle vector) $-\mu$ (non-middle vector)

where λ is the dot product of the vectors other than the middle vector and μ is the dot product of the vectors other than the non-middle vector.

6.6 Jacobi's Identity and Lagrange's Identity

Theorem 6.9 (Jacobi's identity)
For any three vectors $\vec{a},\vec{b},\vec{c}$, we have $\vec{a}\times(\vec{b}\times\vec{c}) + \vec{b}\times(\vec{c}\times\vec{a}) + \vec{c}\times(\vec{a}\times\vec{b}) = \vec{0}$.

Proof

Using vector triple product expansion, we have

$$\vec{a}\times(\vec{b}\times\vec{c}) = (\vec{a}\cdot\vec{c})\vec{b} - (\vec{a}\cdot\vec{b})\vec{c}$$
$$\vec{b}\times(\vec{c}\times\vec{a}) = (\vec{b}\cdot\vec{a})\vec{c} - (\vec{b}\cdot\vec{c})\vec{a}$$

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